Reg. No. | | | | | | | | | | | [17CE45]

FOURTH SEMESTER B.TECH EXAMINATION, JUNE – 2019

STRUCTURAL ANALYSIS-I

Time: 3 Hours **Maximum Marks:70**

Instructions:

i. Answer any **FIVE** full Questions choosing atleast any **TWO** full questions from each part.
ii. Assume any missing data suitably if necessary
iii. Draw neat diagrams wherever required

PART - A

1. a. On the basis of the strain energy principle, derive an expression for the force and the deflection of a cantilever beam carrying a load W at its free end. Calculate the value of the deflection and the strain energy stored in the beam if the load W = 50 kN, Span = 2m, E = 200 kN/mm^2, and I = 5 X 10^{-4} m^4 **07**

b. Determine the deflection at the center BC of a portal frame subjected to UDL of 15 kN/m as shown in Figure (1). EI is constant throughout. **07**

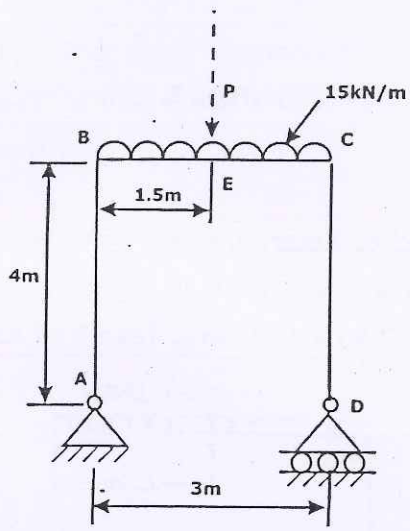

Fig. 1.

2. a. A hollow circular column of external diameter 200mm and thickness 5mm is 6m long. For the material properties of the column, a column is considered short if the slenderness ratio is less than or equal to 70, and long if the ratio is more than 100. For intermediate columns, a linear variation in critical stress from a yield stress of 280 MPa to a proportional limit of 210 MPa may be used. Find the load capacity of the column. **07**

b. Determine the Euler load for the column section shown in Figure (2) if its length is 3m and **07**

i. If its ends are hinged and
ii. If its ends are fixed. E = 200 GPa.

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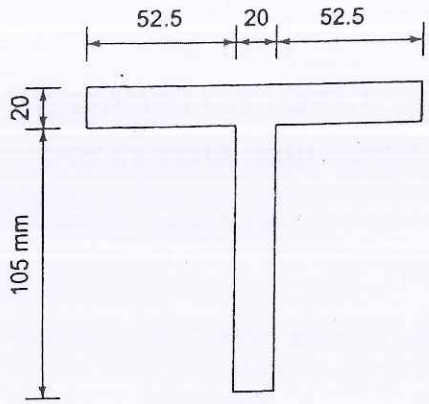

Fig. 2.

3. An overhanging beam is loaded as shown in Figure (3). Find the deflection under the loads. EI is constant. (All distances are in m and ordinated in kNm). Also draw BMD and SFD. Use Moment Area Method. **14**

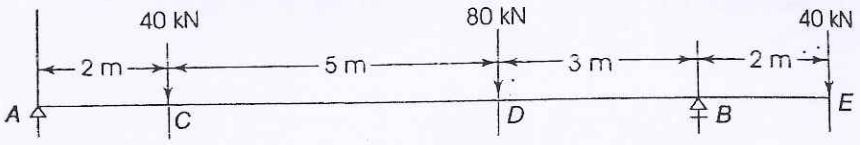

Fig. 3.

4. Use virtual work method to determine the deflection at the mid-span of the beam of a symmetrical frame shown in Figure (4). I is a constant for all members and is equal to 650 X 10^6 mm^4. Take E = 200 kN/mm^2. **14**

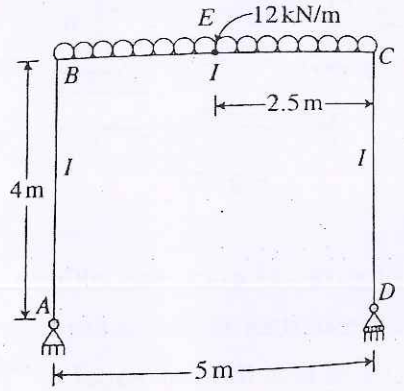

Fig 4.

5. Analyse the continuous beam shown in Figure (5) and draw the SFD and the BMD. EI is constant. Use 3 Moment theorm. **14**

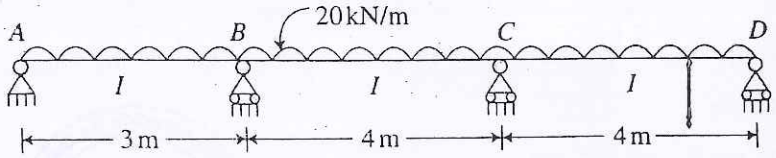

Fig 5.

6. A set of concentrated loads is shown in Figure (6). find the absolute maximum moment in a simple beam of span 8m. **14**

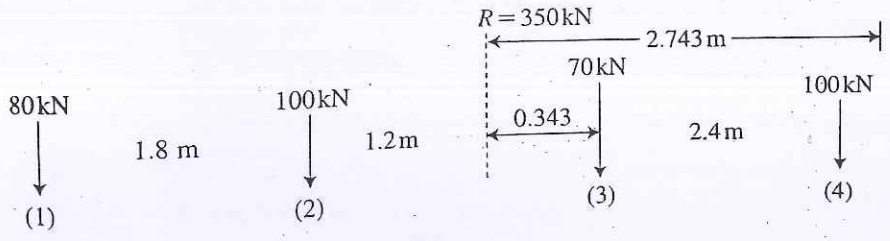

Fig 6.

7. A symmetrical three hinged circular arch has a span of 16m and a rise to the central hinge of 4m. it carries a vertical load of 16 kN at 4m from left hand end. Find (a) the magnitude of the thrust at the springings, (b) the reactions at the supports, (c) bending moment of 6m from left hand hinge, and (d) the maximum positive and negative bending moment. **14**

8. A light cable, 18m long, is supported at two ends at the same level. The supports are 16m apart. The cable supports three loads of 8, 10, 12 N dividing the 16m distance in four equal parts. Find the shape of the string and tension in various portions. **14**